Intelligent Mode Selection for Dual-Functional Radar-Communication in Autonomous Vehicles

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BTP Review 1

Presentation Outline

Introduction: Why Autonomous Vehicles Need Both Radar and Communication

Autonomous Vehicles (AVs) rely on two distinct but hardware-similar subsystems:

- **Radar (Sensing):** Detects obstacles, pedestrians, and hazards in real time. Critical for collision avoidance, especially in poor visibility (rain, fog, night).
- **Communication (V2X/V2I):** Transmits sensor data, traffic status, and telemetry to edge/cloud infrastructure or nearby vehicles for cooperative driving, route planning, and remote monitoring.

The core tension:

- Both functions increasingly share the same RF front-end, spectrum band, and hardware (mmWave radios also make good radars).
- An AV cannot always do both at once with limited hardware — every time slot spent communicating is a time slot *not* spent sensing, and vice versa.
- This shared-resource conflict motivates **Dual-Functional Radar-Communication (DFRC)** systems.

Evolution: From Separate Systems to Joint Radar-Communication

1. **Separate Systems (Traditional):** Dedicated radar hardware + dedicated communication radio, operating on different spectrum. Simple, but costly, bulky, and spectrum-inefficient as vehicular spectrum demand grows.
2. **Spectrum Congestion Pressure:** Growing V2X traffic and radar density (multiple AVs, mmWave bands) pushed research toward **sharing** hardware/spectrum instead of duplicating it.
3. **Joint Radar-Communication (JRC) / DFRC:** A single hardware platform performs both functions, saving cost, size, and power — the foundation of modern automotive DFRC research.
4. **Fixed / Scheduled Sharing (Early DFRC):** Static time-cycle division or preamble-based reservation (e.g., 802.11ad-style schemes) — radar and comm get a pre-agreed fixed share of time, regardless of actual need.
5. **Adaptive, Learned Sharing (Current Direction):** Since road conditions, traffic, and weather change dynamically, a **learned, adaptive policy** decides mode selection in real time — this is the direction our base paper (iRDRC) takes.

Why Fixed Scheduling Falls Short

Static/scheduled resource-sharing assumes a predictable world. Reality is not predictable:

- **Weather and road conditions** change continuously — a fixed radar/comm ratio that works on a clear highway is unsafe in fog or on a wet, winding road.
- **Traffic and moving-object density** varies — more radar time is needed near pedestrians/intersections; more comm time is fine on an empty highway.
- **Channel quality** fluctuates — a fixed schedule wastes comm slots on a bad channel and under-uses a good one.
- Fixed schemes cannot trade off **throughput vs. safety** on the fly — they are tuned once, offline, for an average case.

Consequence: The system either wastes capacity (over-cautious radar use) or risks missed hazards (over-aggressive comm use). This motivates replacing fixed rules with a **learned, state-aware decision policy**.

Toward Intelligent, Adaptive DFRC

The shift in this line of research:

- Treat mode selection (radar vs. communication) as a **sequential decision-making problem**, not a fixed schedule.
- At every time step, the AV observes its situation — queue backlog, channel quality, road/weather/speed/traffic conditions — and **decides** which mode to use.
- This naturally maps to a **Markov Decision Process (MDP)**, solved using reinforcement learning (e.g., Deep Q-Networks).

This is exactly the approach of our base paper:

iRDRC — An Intelligent Real-Time Dual-Functional Radar-Communication System for Automotive Vehicles (Hieu et al., IEEE Wireless Communications Letters, 2020)

Next: a closer look at this base work, followed by recent papers that build on the same idea.

Literature Review: Base Work — iRDRC (Hieu et al., 2020)

iRDRC models the AV's radar/communication choice as an **MDP** $\langle S, A, R, P, \gamma \rangle$:

- **State:** (d, c, r, w, v, m) — data queue length, channel quality, and four binary environmental risk factors (road, weather, speed, moving objects); 352 discrete states.
- **Action:** Binary choice each time step — communicate ($a = 0$) or scan with radar ($a = 1$).
- **Reward:** Safety-first design — small positive reward for successful communication, a large penalty (-50) for missing a hazard while communicating, and a risk-scaled reward for successful radar detection.
- **Solver:** Deep Q-Network (DQN) learns the optimal policy $\pi^*(s)$ from experience, without needing a hand-crafted schedule.

Significance: First to frame automotive DFRC mode-selection as a *learned* policy rather than a fixed/static allocation — the direct foundation for this project.

Literature Review: Recent Advances Building on This Idea

Reference	Approach	Modification / Extension over iRDRC-style single-AV DQN
Lee et al. (2022), *IEEE TVT*	Graph Neural Network (GNN) for JRC	Moves from a single-AV binary decision to a **multi-vehicle cooperative** setting; needs **minimal prior system-model knowledge**; introduces a **data-usefulness metric** to decide what/whom to transmit to, with a tunable radar-comm performance balance.
Li et al. (2024), *IEEE (THz-JSAC)*	Dynamic GNN for THz sensing + comm	Extends to **Terahertz-band** joint sensing-communication across a **vehicular network** (not one AV); each vehicle can offer sensing-only, comm-only, or **both**; a topology-adaptive dynamic GNN outperforms static GNN/optimization baselines by 17–28%.
IEEE TVT (2024)	Hybrid DDPG + DQN for ISAC resource allocation	Replaces the single binary DQN action with a **hybrid continuous + discrete** action space: DDPG allocates channel/power (continuous), DQN allocates bandwidth (discrete), jointly across **many vehicles** under spectrum scarcity.

Common trend: All three move from iRDRC's *single-AV, binary-action* MDP toward *multi-vehicle, richer-action* formulations (GNN-based coordination, continuous resource allocation) — pointing to where the open research space likely lies.

Research Gap: Setting the Stage

The base approach (iRDRC): At every instant, the vehicle must choose between using its sensor (radar) or its communication link — not both at the same time. The decision follows a clear, safety-first rule: prioritize sensing when conditions are risky, and prioritize communication when conditions are safe. This works well for **a single vehicle operating independently**.

Subsequent research extended this idea in three directions:

- Allowing **multiple vehicles to cooperate** and share information.
- Allowing a vehicle to **perform both functions simultaneously**, rather than strictly choosing one.
- Allowing **finer control** over how much attention is given to each function.

Each of these represents genuine progress in flexibility. However, a closer look at this progression reveals an important trade-off.

Research Gap: The Core Limitation

Four limitations remain consistent across the base work and its extensions:

- **An unrealistic knowledge assumption.** Every method assumes the vehicle already knows its channel quality and risk conditions at each step — in practice this must be estimated from noisy signals.
- **Sensing treated as detection only.** Radar is always modeled purely as a hazard-detection tool; none of these methods recognize that sensing also *resolves the vehicle's own uncertainty*.
- **A diluted safety priority.** As systems became more flexible (cooperative, multi-level), the strict safety-first rule was replaced with a tunable balance — safety became one objective among several, not the dominant priority it was meant to be.
- **Communication constraints are ignored.** Packet deadlines, packet priority, and quality-of-service are absent from every model reviewed — no analytical or learned V2X-ISAC formulation treats communication as anything more than undifferentiated throughput.

The Gap

No existing work models decisions under realistic uncertainty — where sensing is *both* a hazard-detection and uncertainty-reducing action, and communication is deadline- and priority-aware — while preserving a strict, safety-dominant priority instead of a soft trade-off.

Problem Statement

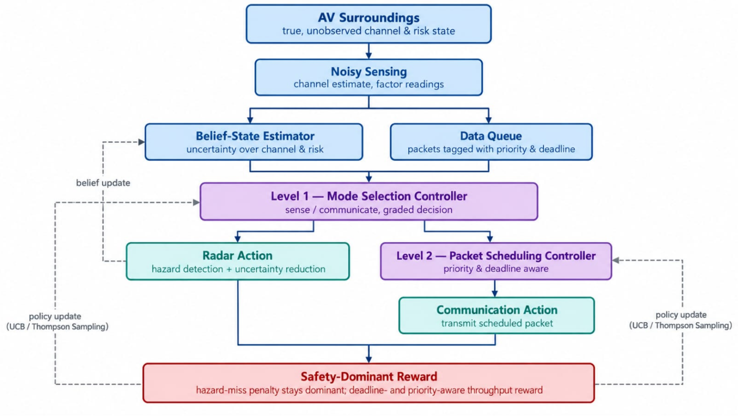
Limitations of Existing Approaches:

- Assume the vehicle already knows its channel and surroundings perfectly
  - In reality, this must be sensed under uncertainty
- Treat radar as all-or-nothing hazard detection only
  - Ignore that sensing can also reduce uncertainty
- Treat every data packet as equally urgent
  - Ignore deadlines, priority, and quality-of-service

Proposed Direction:

- Model the decision under **realistic uncertainty**
- Let sensing serve a **dual role** — detect hazards *and* reduce uncertainty
- Make scheduling **deadline- and priority-aware**
- Keep **safety as the top priority**, always

Proposed System Architecture



Proposed Work: Research Methodology — Architecture Explained

- **AV Surroundings → Noisy Sensing**
  - True channel & risk state is never observed directly, only sensed with noise
- **Belief-State Estimator**
  - Maintains an uncertainty-aware estimate instead of assuming perfect knowledge
- **Data Queue**
  - Packets are now tagged with priority and deadline, not just counted
- **Level 1 — Mode Selection Controller**
  - Decides sense vs. communicate as a graded decision, using belief + queue as context
- **Radar Action**
  - Dual role: detects hazards *and* feeds a belief update back to the estimator
- **Level 2 — Packet Scheduling Controller**
  - When communicating, picks which packet to send based on priority & deadline
- **Safety-Dominant Reward**
  - Hazard-miss penalty stays dominant; trains both controllers via UCB / Thompson Sampling

Base Paper: MDP Formulation (Sections II–III)

The MDP tuple: $\langle S, A, R, P \rangle$

- **State space** S (Eq. 2): (d, c, r, w, v, m), $d \in \{0, \ldots, 10\}$, others binary $\rightarrow$ **352 states**
- **Action space** A: $a = 0$ (communicate) or $a = 1$ (radar)
- **Environment model** (Eq. 1): $P(\oplus) = \sum_i \left[\tau_i p_0^i + (1 - \tau_i) p_1^i\right]$, $i \in \{r, w, v, m\}$
- **Reward function** (Eq. 3): $+r_1/+r_2$ (comm, no event, good/bad channel); $-r_3$ (comm, event); $+r_4(b+1)$ (radar, event); 0 (radar, no event)
- **Objective** (Eq. 4): $\max_{\pi} G(\pi) = \mathbb{E}\Big\{ \sum_t \gamma^t r_{t+1}(\pi) \Big\}$

Scope of this review: Implemented in full — state space, action space, environment model, reward function, and objective — everything through the end of Section III. Section IV (the DQN algorithm) is Phase 3.

Phase 1 Results: Environment Verification

Sanity checks (MATLAB implementation):

- State space size: **352** (matches Eq. 2 exactly)
- $P(\oplus)$ computed from Eq. 1: **0.1002**

Three fixed policies compared ($T = 500$ steps, no learning yet):

Policy	Total Reward	Discounted G	Mean d	Events
Round-robin	-302.00	-42.19	1.58	48/500
Always-comm	-2044.00	-546.83	0.96	58/500
Always-radar	$+595.00$	$+105.75$	9.90	49/500

Inference: Always-radar avoids the $-r_3$ penalty entirely but leaves the queue saturated (near-zero throughput). Always-comm drains the queue but repeatedly eats the -50 penalty. Round-robin sits between the two — exactly the throughput/safety trade-off the reward function is designed to expose, and exactly what a learned policy (Phase 3) must resolve.

Requisites Analysis and Project Plan

Phase	Focus
Phase 1 *(Completed)*	Part 1 implementation of the base paper: state space (Eq. 2), action space, environment model / unexpected-event probability (Eq. 1), reward function (Eq. 3), and discounted-return objective (Eq. 4) — the full simulation environment, up to but not including the DQN algorithm (Section IV).
Phase 2	Complete the base paper by implementing the DQN algorithm (Section IV) and reproducing the iRDRC baseline results (Section V); then stress-test it across many scenarios — varying channel/event probabilities, reward weights, and environmental conditions — to empirically identify its drawbacks: the perfect-knowledge assumption breaking under estimation noise, the rigidity of the binary action, and the absence of QoS-aware scheduling.
Phase 3	Halfway implementation of the proposed work: belief-state estimator, dual-role radar action, and the Level 1 Mode Selection Controller; evaluate the initial improvement over the base paper's DQN baseline from Phase 2.
Phase 4	Complete the proposed architecture — Level 2 Packet Scheduling Controller, deadline-/priority-aware reward, and full hierarchical training (UCB / Thompson Sampling); run the full comparative evaluation and ablation studies against the base paper and the literature reviewed; consolidate results, write the thesis, and prepare the final defense.

Expected Outcomes

- A validated uncertainty-aware formulation — a belief-state estimator producing a usable channel/risk estimate instead of assuming perfect knowledge, with radar confirmed to serve a **dual role**: hazard detection *and* measurable uncertainty reduction.
- Demonstrated benefit of a **graded, multi-level** mode-selection space over iRDRC's strict binary action.
- A learned hierarchical policy (mode-selection + scheduling controllers) that **outperforms all three Phase 2 baselines** in discounted return $G(\pi)$ — exceeding even always-radar's safety-only ceiling ($G = +105.75$) — while keeping **safety dominant throughout** and maintaining meaningful throughput.
- Reduced miss-detection probability vs. the iRDRC baseline, without sacrificing throughput the way always-radar does; faster convergence than a flat DQN over the same (or richer) state space.
- A working deadline- and priority-aware packet scheduler, with measurably better on-time delivery for high-priority packets than iRDRC's undifferentiated throughput model.
- A direct, quantitative comparison against the base paper and the three recent extensions reviewed, positioning this work's contribution within the current literature.

References

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Thank You

Questions & Discussion